SECTION A (25 marks)
This section consists of **3** questions. Answer **all** questions.

1. Evaluate the following limits:

   (a) $\lim_{x \to -1} \dfrac{x^3+1}{x+1}$ [3 marks]

   (b) $\lim_{x \to 3} \dfrac{\sqrt{x+1}-2}{x-3}$ [3 marks]

   (c) $\lim_{x \to \infty} \dfrac{(2-x)(x-1)}{(x-3)^2}$ [3 marks]

2. (a) Find $\dfrac{dy}{dx}$ if:

   (i) $y = \ln[\sin(9-2x)]$. [4 marks]

   (ii) $y = \tan(e^{2x})$. [3 marks]

   (b) Evaluate $\dfrac{dy}{dx}$ for $y = (x^2+1)^5$ when $x = -1$. [4 marks]

3. Given a function $f(x) = 2x^4 - 16x^2 + 14$. By using the second derivative test, find the maximum and the minimum points of $f(x)$. [5 marks]

SECTION B (75 marks)
This section consists of 7 questions. Answer **all** questions.

1. Given $z_1 = 2+3i$ and $z_2 = 4-4i$. Express $z = \dfrac{(\overline{z_2})}{(\overline{z_1})} + \left[\left(\dfrac{i^3}{-z_2}\right)\right]$ in Cartesian form. Hence, find the modulus, argument and polar form for z.

[10 marks]

2. Find the solution set of the inequality $\dfrac{1}{|x-1|} \le \dfrac{2}{|x+2|}$.

[10 marks]

3. If given $g(x) = \dfrac{x+a}{bx+c}$ and $g^{-1}(x) = 1 + \dfrac{3}{x+1}$, find the values of the constants a , b and c .

[6 marks]

4. The function f and g are defined as

$$f(x) = 3\sqrt{1-x}$$

$$g(x) = \ln\left(\frac{x-2}{3}\right)$$

(a) If the function $h(x)$ is defined as $h(x) = \begin{cases} f(x) \\ g(x) \end{cases}$ for $x \in R$, sketch $h(x)$ and state the domain and range of $h(x)$.

[4 marks]

(b) Show that $h(x)$ is not a one to one function.

[2 marks]

(c) Find $g^{-1}(x)$ and hence, sketch $g^{-1}(x)$ on the same axes with $g(x)$. Hence, state the domain and range of $g^{-1}(x)$.

[6 marks]

5. Consider the function:

$$f(x)=\begin{cases} \dfrac{x}{x-3} & ; \quad x<-3 \\ x+1 & ; \quad -3\le x\le 3 \\ (kx-3)^2 & ; \quad x>3 \end{cases}$$

Where $k>1$.

(a) Determine whether $f(x)$ is differentiable at $x=-3$. [5 marks]

(b) Determine the value of k if the function $f(x)$ is continuous at $x=3$. [4 marks]

(c) Find the horizontal asymptote of the graph of $f(x)$ as $x\to-\infty$. [2 marks]

6. (a) Parametric equation of a curve is given by $x=t^3-3t$ and $y=3t^2$.
Find

(i) $\dfrac{dy}{dx}$ in terms of t. [4 marks]

(ii) $\dfrac{d^2y}{dx^2}$ when $t=\sqrt{3}$. [4 marks]

(b) Find the values of $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$ at the point $(1,3)$ on the curve $3x^2+y^2=4y$. [8 marks]

7. Salt water is poured into a right inverted cone with a semi-vertical angle of 45° at a rate of $3\pi\,\text{cm}^3\text{s}^{-1}$. Find

(a) The depth of the salt water after 45 seconds. [7 marks]

(b) The rate of change of its depth at that instant. [3 marks]

END OF QUESTION

Final Answer:

SECTION A

1. (a) 3 (b) $\frac{1}{4}$ (c) -1

2. (a) (i) $-2\cot(9-2x)$ (ii) $2e^{2x}\sec^2\left(e^{2x}\right)$ (b) -160

3. $(0,14)$ maximum point ; $(2,-18)$, $(-2,-18)$ minimum points

SECTION B

1. $z=\frac{147}{104}+\frac{45}{104}i$; $|z|=\sqrt{\frac{909}{416}}$; $\text{Arg}(z)=0.2971\,rad$;

   $z=\sqrt{\frac{909}{416}}\left[\cos(0.2971\,\text{rad})+i\sin(0.2971\,\text{rad})\right]$

2. $(-\infty,0]\cup[4,\infty)$, $x\neq-2$

3. $a=-4$, $b=-1$, $c=1$

4. (a) $D_h=(-\infty,1]\cup(2,\infty)$; $R_h=(-\infty,\infty)$

   (b) by using horizontal line test

   (c) $g^{-1}(x)=3e^x+2$; $D_{g^{-1}}=(-\infty,\infty)$; $R_{g^{-1}}=(2,\infty)$

5. (a) $f(x)$ is not differentiable at $x=-3$.

   (b) $k=\frac{5}{3}$

   (c) $y=1$

6. (a) (i) $\frac{dy}{dx}=\frac{2t}{t^2-1}$; $\frac{d^2y}{dx^2}=\frac{-2-2t^2}{3(t^2-1)}$; $\left.\frac{d^2y}{dx^2}\right|_{t=1}=-\frac{1}{3}$

   (b) $\frac{dy}{dx}=\frac{3x}{2-y}$; $\left.\frac{dy}{dx}\right|_{(1,3)}=\frac{3(1)}{2-3}=-3$; $\left.\frac{d^2y}{dx^2}\right|_{(1,3)}=-12$

7. (a) $h=7.3986$ cm

   (b) $\left.\frac{dh}{dt}\right|_{h=7.3986}=0.0548$ cm/s